BDA400 — Data Science Tools and Techniques

Assignment 5: Technical Analysis Using R Development Phase

Student: Hanin Helfawi

Submission file: Helfawi_Hanin_CA_BDA400_A05.pdf

Repository: https://github.com/hanin1987h/markdown-buddy-hanin-helfawi

Assignment folder: BDA400_Assignment5/

This document explains the nine base-R indicator implementations and how to load and test them. The executable functions are separate .R files in the repository.

1. Development approach

Each function uses core R operations rather than third-party technical-analysis libraries. The implementation follows the assignment function signatures and the supplied calculation descriptions. Inputs are checked where appropriate, and the functions return numeric vectors, named lists, or crossover labels.

sma.R — Simple Moving Average

sma(data, period) calculates the arithmetic mean of each consecutive window of period observations. It returns a vector of length n - period + 1. The function checks that the period is a positive integer and the dataset has sufficient observations.

ema.R — Exponential Moving Average

ema(data, period) applies a multiplier of 2/(period + 1), initializes the first EMA at the first price, and recursively updates later values. The returned numeric vector has the same length as the input.

macd.R — Moving Average Convergence Divergence

macd(data, short_period, long_period, signal_period) calls ema() twice, subtracts long EMA from short EMA, computes an EMA of the resulting MACD line, and subtracts signal from MACD to obtain the histogram. It returns macd_line, signal_line, and histogram.

stdev.R — Population Standard Deviation

stdev(data) calculates the mean, squared deviations, population variance (sum of squared deviations divided by n), and square root of the variance. This follows the population standard-deviation formula specified in the assignment.

linreg.R — Linear Regression

linreg(regressionSource, regressionLength, regressionOffset) selects the specified trailing interval with the assignment offset logic, calculates least-squares slope and intercept using index values, and returns slope, intercept, and predicted_values. The selected interval must have at least two observations.

rsi.R — Relative Strength Index

rsi(data, period) calculates consecutive price changes, separates gains from losses, seeds average gains/losses with the first period of changes, and uses Wilder smoothing for later RSI values. Initial unavailable entries are NA. Zero-loss cases are handled explicitly.

stoch_rsi.R — Stochastic RSI

stoch_rsi(data, period, k_period, d_period) calls rsi(), removes initial unavailable values, normalizes valid RSI values between zero and one using their minimum and maximum, then applies sma() to calculate k_line and d_line. Constant RSI values are assigned a normalized value of zero.

crossover.R — Crossover

crossover(arr1, arr2) compares adjacent points of equal-length numeric arrays. It labels upward crossings Up, downward crossings Down, and other positions None, following the supplied pseudocode.

crossunder.R — Crossunder

crossunder(arr1, arr2) identifies points where the first series moves from greater than or equal to the second series to strictly below it. It returns True or False as character strings, with None at the first position.

2. How to run the code

1. Open R or RStudio. Download the nine .R files from the BDA400_Assignment5 repository folder into one local folder. Set the R working directory to that folder.

2. Load the files in dependency order: sma.R, ema.R, stdev.R, linreg.R, rsi.R, crossover.R, crossunder.R, macd.R, and stoch_rsi.R. In R, use source("sma.R"), source("ema.R"), and so on.

3. Run sample calls, such as sma(c(10,12,15,20), 3), ema(c(10,12,15,20), 3), macd(c(100,105,110,115,120,125,130),3,5,2), and stdev(c(10,12,15,20)).

4. Run the separate test_indicators.R script if included in the repository; it performs basic expected-value and output-shape checks.

3. Notes on testing and limitations

The implementation has not been executed in an R runtime as part of preparing this document. Therefore, these descriptions are not a claim of successful runtime testing. Before submission, run all files and tests in RStudio and review the outputs.

The assignment provides slightly inconsistent descriptions for some indicators. The StochRSI implementation follows the supplied pseudocode using a global minimum/maximum over valid RSI values, rather than a rolling min/max. The crossover function follows the Up/Down/None pseudocode, and crossunder returns the specified character labels.

The first regression window depends on the supplied offset. The implementation should be checked for boundary conditions, especially where the selected subset contains fewer than two observations.

4. Repository and submission

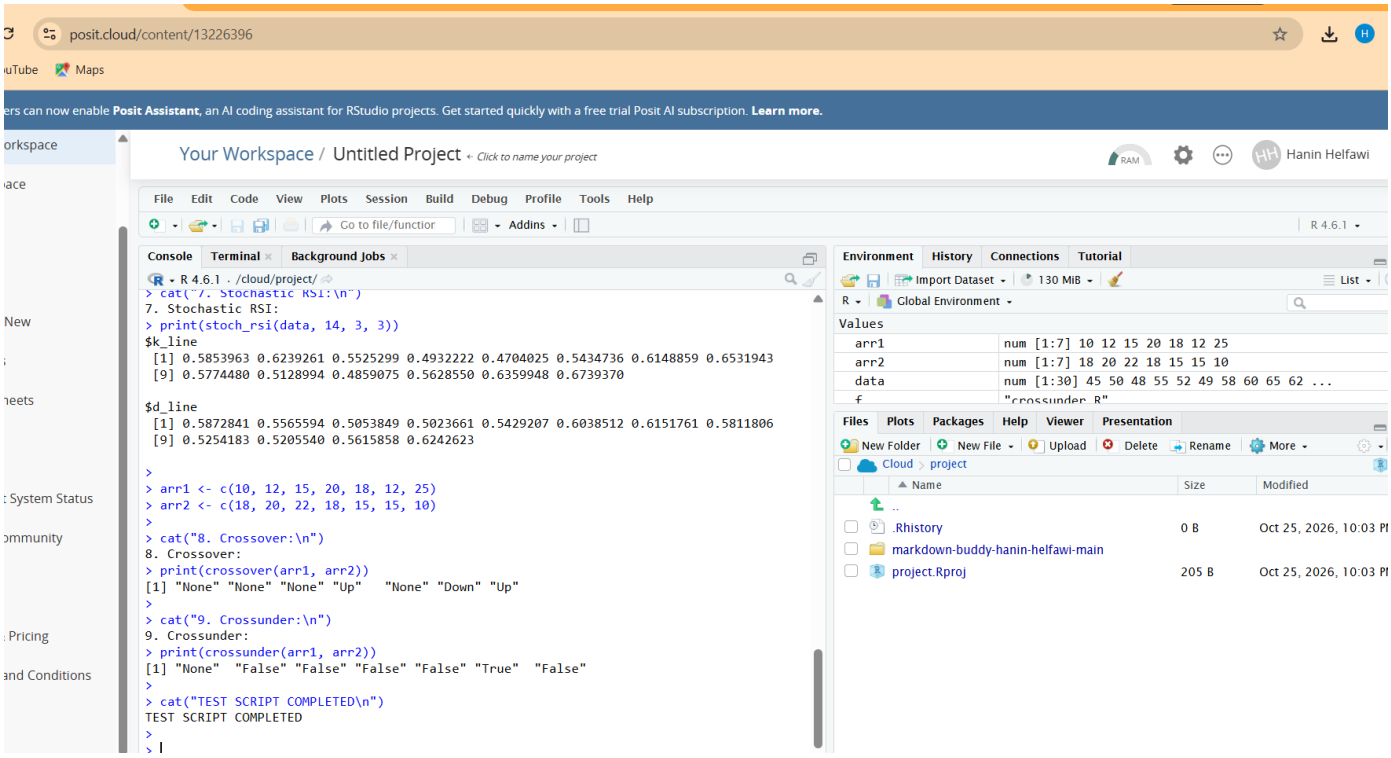
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Submit the requested PDF through the LMS and ensure the repository contains all nine scripts and documentation. Confirm the repository URL is accessible to the instructor and that the code runs before final submission.

4. RStudio execution evidence

The nine R source files were loaded in Posit Cloud without a reported sourcing error. A sample execution script was then run and the console displayed **TEST SCRIPT COMPLETED**. The screenshot below documents visible output for Stochastic RSI, Crossover and Crossunder.



Verification scope: This screenshot demonstrates execution of the sample script, not independent numerical verification of every indicator. Further comparison with reference values and edge-case tests is recommended. The earlier testing note describes the state when the initial report was prepared; this appendix records the later RStudio run.